Geothermal Energy

Picture of Geothermal power plant coming up in Ladakh, 2021

- Earth contains large amounts of geothermal energy with temperature as high as 4400 °C
- This energy comes from
  - magma, molten rock material beneath the surface of the earth or
  - radioactive decay of thorium, potassium and uranium dispersed throughout the earth's interior
- some regions of the earth this molten material sometimes breaks through the earth's crust and produces volcanoes

  In other regions, the hot material is close enough to the earth's surface to heat the underground water trapped by impermeable rock and form steam – appear as Geysers and hot springs on surface
- In such areas geothermal energy is tapped by drilling wells to obtain steam

- Large scope for utilization of geothermal energy
  - to uplift the socio-economic status and the life style of the people, particularly those living in far remote areas of the country and
  - where this source of geothermal energy is in abundance.
- Besides space heating and power generation, the energy could be utilized:
  - In small and cottage industries
  - Drying and processing of conventional and cash crops
  - Animal husbandry, dairy, poultry and fishery development
  - Silviculture
  - Spinning, weaving, painting and garment industry
  - Hard and soft board manufacturing and pulp making
  - Brewing of low alcoholic beverages

- Geothermal Sources are Classified Based on:

  (1) Temperature

  (2) Physical State of H_2O (i.e. water or steam) and

  (3) Type of Energy Usage

- Primary Classification is Resource Temperature:
  - Low Temperature Reservoir: 50-200 °F (10-94 °C)
  - High Temperature Reservoir: >200 °F

What Makes a Good Geothermal Reservoir for Generating Electricity?

- Hot Geothermal Fluids Near Surface (<1-2 mi.)
  - Preferably in Excess of 300°F, but Electrical Generation Is Occurring at Temps. In the Low 200's°F.
- Proximity to Population Base
- Low Mineral and Gas Content
- Location on Private Land
- Proximity to Transmission Lines

World Scenario

Total installed capacity for Geothermal Power is around **13.5 GW**. Leading countries in geothermal power generation capacity are

- USA (3600 MW)
- Philippines (1900MW)
- Indonesia (1600 MW)
- New Zealand (1000 MW)
- Mexico (900 MW)
- Italy (800 MW)
- Turkey (800MW)
- Iceland (700 MW)
- Kenya (600 MW)
- Japan (500 MW).

Total installed capacity for geothermal direct heat utilization for heating/cooling (excluding heat pumps) is around **23 GWth**.

Leading countries in geothermal direct heat use are
China (6.1 GWth), Turkey (2.9 GWth), Japan (2.1 GWth), Iceland (2.0 GWth) & Italy (1.4 GWth).
Total installed capacity for Ground Source Heat Pump (GSHP) is around **50.3 GWth** with leading markets as USA, China & Europe (France, Germany, Italy & Sweden).

Indian Scenario

- India - still at nascent stage of geothermal energy utilization with no geothermal power plant set up in the country so far due to
    - high upfront cost of Rs 30 Cr/MW
    - indicative Tariff in range of Rs 10 per KWh
    - site specific deployment
    - lack of load center
    - power evacuation facility nearby
    - high risk involved in exploration
- Geological Survey of India (GSI) with CSIR - National Geo-physical Research Institute (NGRI) carried out preliminary resource assessment for the exploration and utilization of geothermal resources in 1970s & 1980s in the country
- As per preliminary investigations undertaken by the GSI,
    - there are around 300 geothermal hot springs in India
    - Most of these geothermal hot springs are in medium potential (100 C to 200 C) and low potential (<100 C) zones.
    - Promising geothermal sites for electric power generation are Puga Valley & Chummathang in Jammu & Kashmir, Cambay in Gujarat, Tattapani in Chattisgarh, Khammam in Telangana & Ratnagiri in Maharashtra

    - Promising geothermal sites for direct heat use applications are Rajgir in Bihar, Manikaran in Himachal Pradesh, Surajkund in Jharkhand, Tapoban in Uttarakhand & Sohana region in Haryana.

Technology for Power Generation

Hot water and steam from deep underground can be piped-up through underground wells and used to generate electricity in a power plant

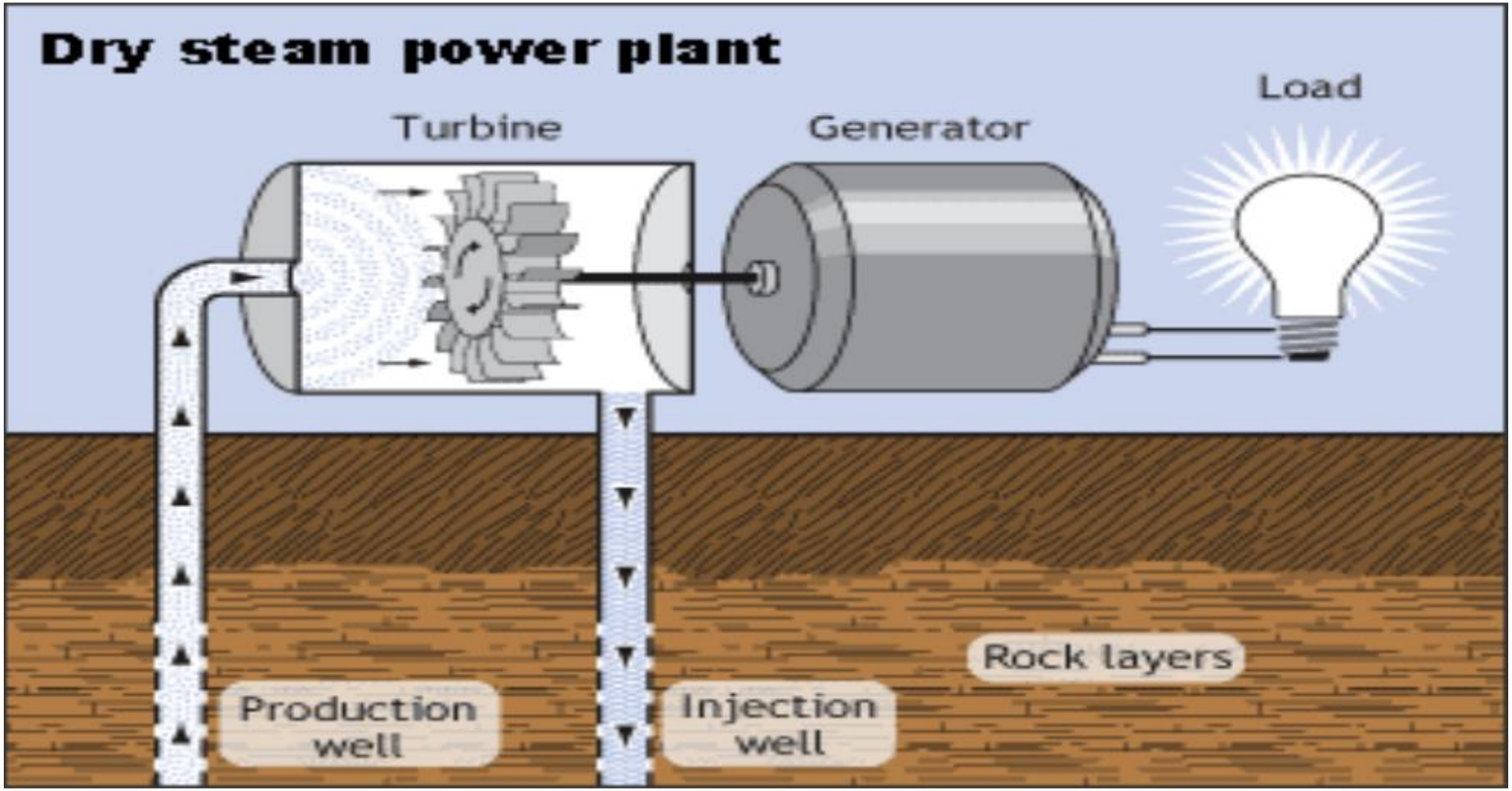
Three types of geothermal power plants:

i) Dry Steam Plants which use geothermal steam directly

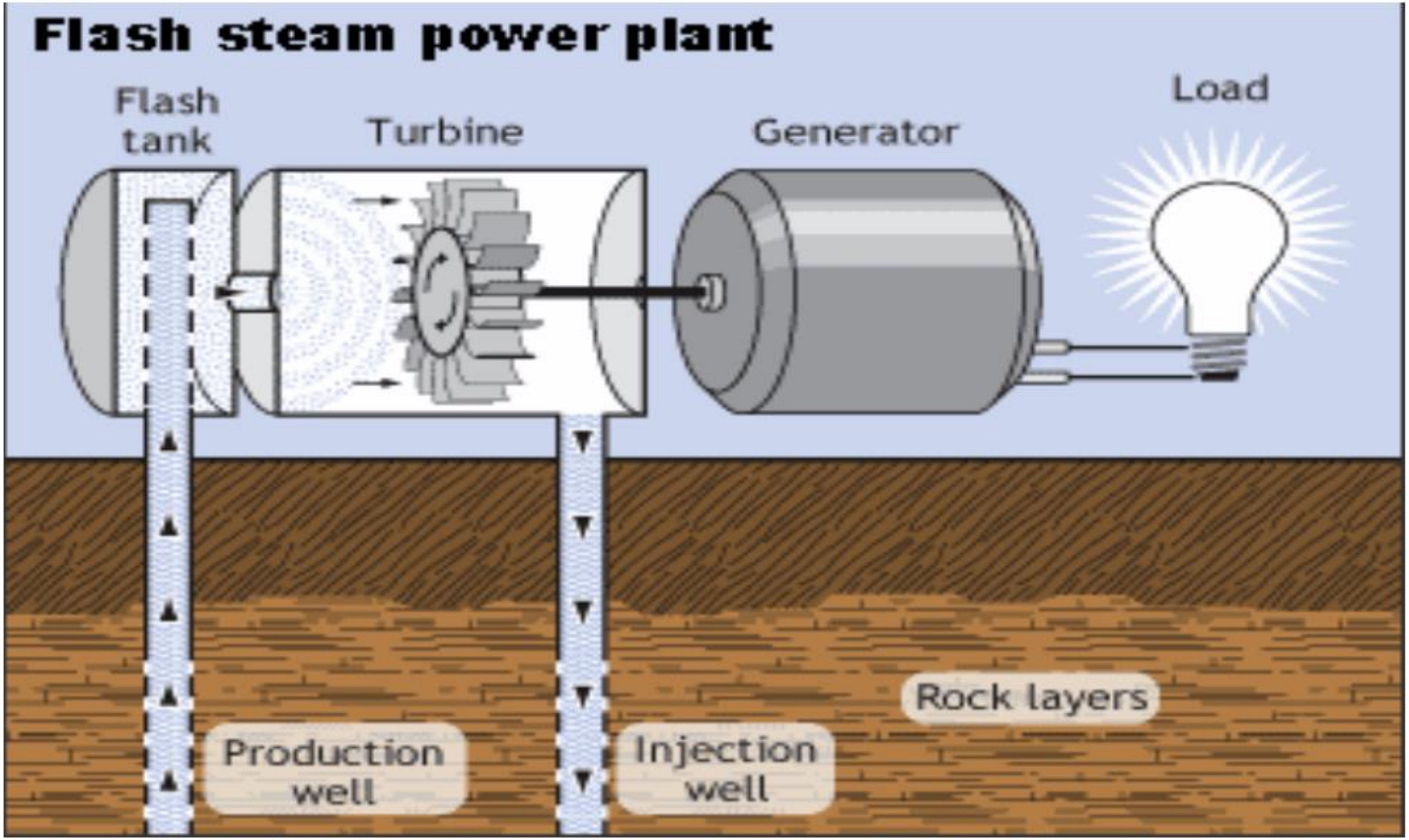
- Dry steam power plants use very hot (>235 °C) steam from the geothermal reservoir
- The steam goes directly through a pipe to a turbine to spin a generator that produces electricity

ii) Flash Steam Plants

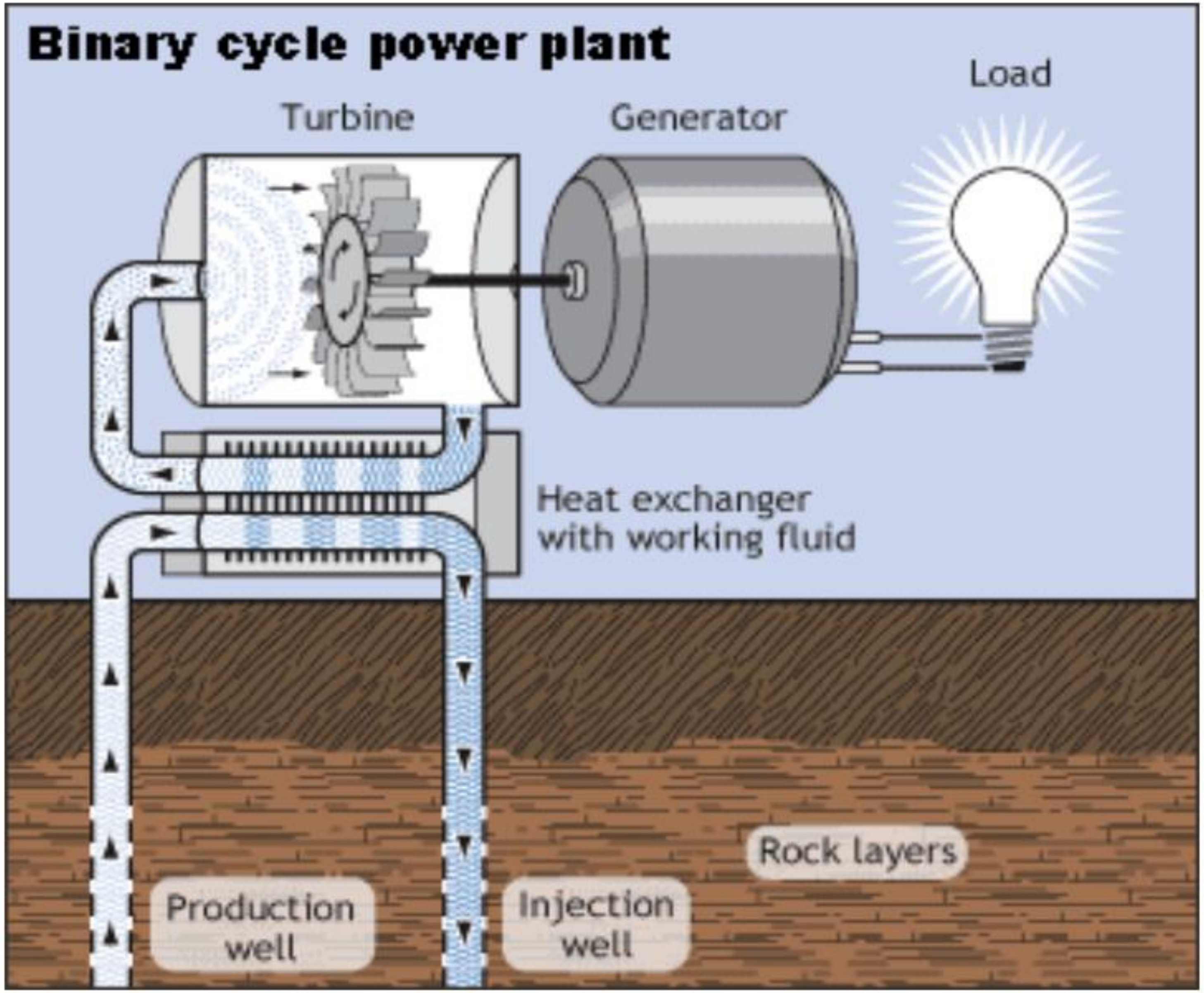
- use high pressure hot water to produce steam
- Flash steam power plant use hot water (>182 ºC) from the geothermal reservoir
- When the water is pumped to the generator, it is released from the pressure of the deep reservoir.
- Sudden drop in pressure causes some of the water to vaporize to steam, which spins a turbine to generate electricity.
- Hot water not flashed into steam is returned to the geothermal reservoir through injection wells



Dry steam geothermal power plant (Source: US Department of Energy)



Flash steam geothermal power plant (Source: US Department of Energy)



Binary cycle geothermal power plant
(Source: US Department of Energy)

(iii) Binary Cycle Plants

- Use moderate temperature water (107 to 182 ºC) from the geothermal reservoir
- In binary systems, hot geothermal fluids are passed through one side of a heat exchanger to heat a working fluid in a separate adjacent pipe.
- The working fluid, usually an organic compound with a low boiling point such as Iso-butane or Iso-pentane, is vaporized and passed through a turbine to generate electricity.

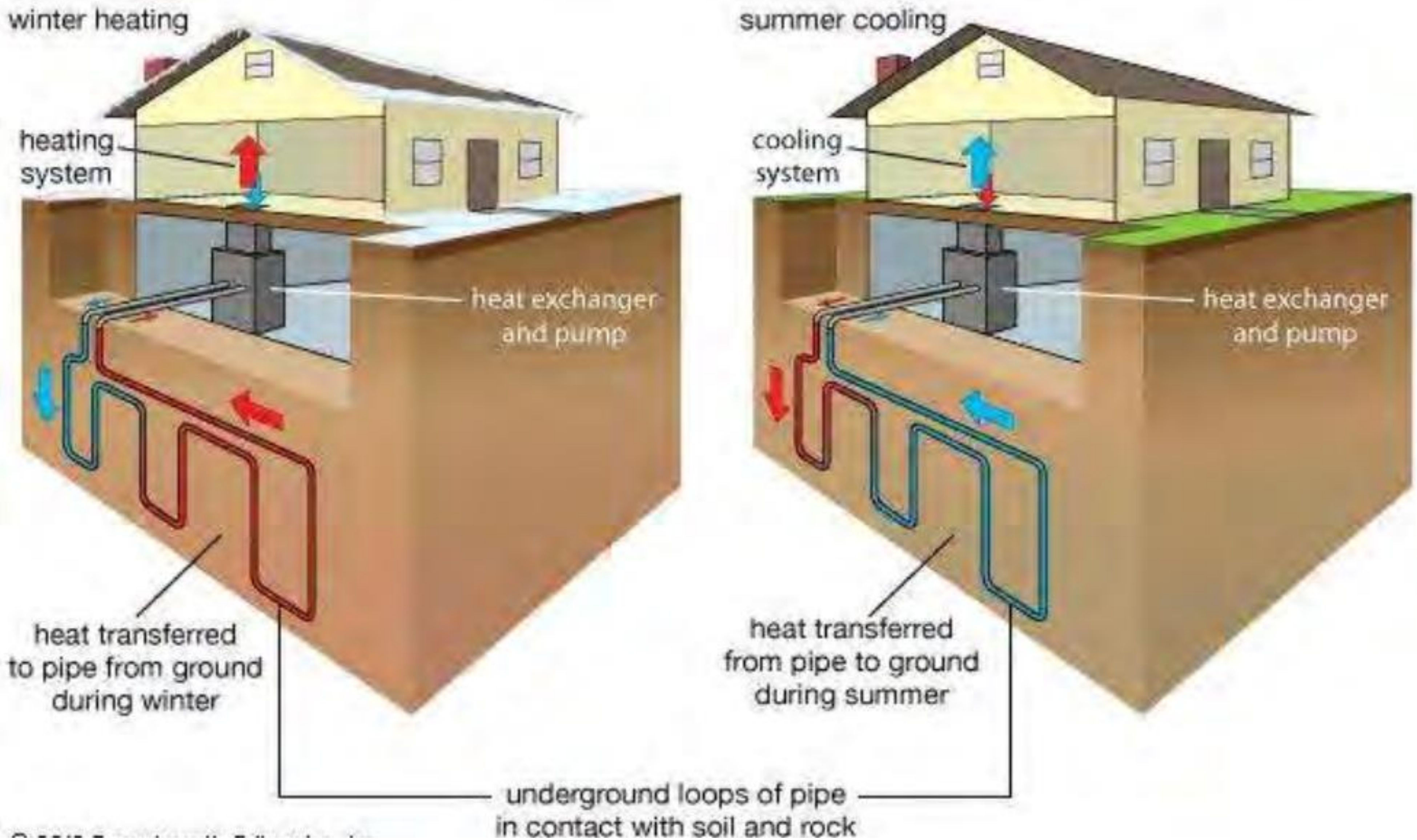
Other Thermal Applications

i) Space/District Heating:

- In areas where hot springs or geothermal reservoirs are **near the Earth's surface**, hot water can be piped in directly to heat homes or office buildings.
- Geothermal water is pumped through a heat exchanger, which transfers the heat from the water into the building's heating system.
- The used water is injected back down a well into the reservoir to be reheated and used again.
- **ii) Geothermal Heat Pump/ Ground Source Heat Pumps:**
    - A few feet under the ground, the soil or water remain a constant 50 to 60 degrees Fahrenheit (10-15 degrees Celsius) year-round.
    - In this method, geothermal heat pumps use a system of buried pipes linked to a heat exchanger and ductwork into buildings.
    - In winter the relatively warm earth transfers heat into the buildings and in summer the buildings transfer heat to the ground or uses some of it to heat water. These heat pumps function as both air - conditioning and heating systems.

      Fluid circulates through a series of pipes under the ground or beneath the water of a pond or lake and into a building.

      An electric compressor and heat exchanger pull the heat from the pipes and send it via a duct system throughout the building. In the summer the process is reversed. The pipes draw heat away from the house and carry it to the ground or water outside, where it is absorbed



Geothermal Heat pump

Benefits of Geothermal power plant

- Little to No Pollution
  - Flash Plants Emit Only Excess Steam
  - Binary Plants Have No Air or Liquid Emissions!
    - Expected to Be Dominant Type in Future
- Emission of Low Quantities of Greenhouse Gasses
- Homegrown
  - Decreases Dependency On Foreign Energy
- Reliability:
  - Plants Have Very Little Down Time - Avg. Availability is 90% or greater
  - 75-80% for Coal and Nuclear Plants

Benefits of Geothermal power plant

- As Opposed to Burning Fossil Fuels, Current Geothermal Use Prevents the Yearly Emission of:
    - 22 MM tons of CO_2
    - 200k tons of SO_2
    - 80k tons of NOx
    - 110k tons of Particulates
- Some Plants Produce Scale Which Is High in Minerals (Zinc and S)
- But, The Minerals are Now Recyclable and Can be Sold For a Profit!
- No Fuel Usage (storage, transfer, disposal, mining)

Benefits of Geothermal power plant

- Minimal Land Use Compared to Other Energy Sources
- Requires 400 M^2 of Land Per GW of Power Over a 30 Year Period

  Compare that to Coal and Nuclear Plants Which Require Land for Plant, Mining for Fuel, Storage of Fuel and Wastes, Etc.

Disadvantages of Geothermal power plant

- Start-up Costs Are High
  - Geothermal Plants Require Significant Capital Expenditures, But the Fuel Is Free
  - Cost - $1,500-$5,000 / Installed kW Depending on Plant Size, Resource Temp. And Chemistry
- Cost Of Power to Consumer
  - Currently, $0.05 to $0.08 / Kwh
  - Needs to Be $0.03 to Be Competitive
- Water can be corrosive to plant pipes, equipment
  - If water not replaced back into reservoir, subsidence can occur
  - How Much Water is Needed? a MW requires 500 gpm @ 300°F; 1400 gpm @ 200°F.
- Some high mineral / metal wastewater and solid waste is produced
- Smelly gasses – H_2S, Ammonia, Boron
- Release of steam and hot water can be noisy